LL RS485 Grind Controller – Instruction Sheet

⚠️ **USE AT YOUR OWN RISK**
This is a **development project**, not a commercial product.
The hardware and firmware are intended for testing and hobbyist experimentation.

INSTALLATION

1. Turn off servo drive (PSU box)
2. Physically mount the box, using the provided nuts, thumb screws & plastic spacer to clamp it.
3. Remove the wire/connector from the original remote control box, plug it into this box.
4. Turn on drive box and make sure the small box is also turned on (no specific order)

FIRST STARTUP

The controller comes preloaded with **default parameters**, tested on the **RT version of the LL SG98 grinder**.
Be cautious when changing settings — most parameters directly affect how the servo drive operates.

1. The box will show "Please reboot drive" to confirm that it has written the new settings. You will now have to reboot the servo drive (the big PSU box).
2. Press **START** to verify that the motor spins in the **correct direction** when in **NORMAL mode** (not GBW mode).
3. Open **Menu 1**, find **"Calibrate"**, and **long press SELECT** to start the calibration.
   The calibration process runs automatically and will stop when finished.

OPERATING MODES

The controller has two primary operating modes:
NORMAL and **GBW (Grind-by-Weight)**.
You can switch between them by **long pressing the SELECT button** while being in either IDLE mode.

IDLE

When idle in normal mode, the display shows:

- target **RPM**
- **Connection status**
- Auto purge setting
- Pressing START will enable the grinder, pressing SELECT shortly will enter the menu.

NORMAL GRINDING MODE

In this mode, the controller automatically detects bean grinding by monitoring motor load.

- After grinding beans (or no interaction occurs for the set time), the grinder will **automatically purge**
- Pressing **SELECT** during operation will always **stop** the motor.
- If **"Purge /w button"** is enabled, pressing **START** triggers a purge cycle; otherwise, it will stop the motor.

PURGE FUNCTION

(PURGE SEQUENCE: *FREE RUNNING → GRINDING → (delay) → STOP → FORWARD → REVERSE*)

Purging helps clear remaining grounds after grinding stops.

Pressing either button will stop the grinder.

You can customize purge behavior under **Menu 2 → Purge Settings**:

- **Enable or disable** automatic purge (after grind detection)
- Adjust **purge delay** and **purge duration**
- Set **forward** and **reverse purge RPMs** (set to 0 to disable this action)
- Enable **purge by button press**
- Fine-tune purge sensitivity (frames and scalars).

GBW – Grind-by-Weight Mode

In this mode, the controller grinds beans until a **target weight** is reached, using a connected Bluetooth scale.

⚖️ **Recommended scale:** *Bookoo Themis Mini*. (Acaia and Felicita Arc work too)

1. Connect a compatible scale (Bluetooth must be enabled).
   The controller will automatically pair with scale.
2. Place the scale under your dosing cup and **fill the hopper with beans**.
3. Select the desired **target weight**.
4. Press **START** to begin GBW grinding.
   The scale will beep once the target is reached.

The controller **self-learns** over time, improving accuracy automatically.

Note: Coffee ground in GBW mode tends to be slightly **finer** than in NORMAL mode, due to bean pressure. Adjust the grinder coarser to combat this. Unfortunately an emptying hopper will change the grind size over time. Keep the hopper partly filled to have the grind size stable.

GbW settings can be customized in the menu.

To exit GbW mode and return to normal operation, **long press SELECT**.
(This also disconnects the scale.)

MENU

Enter the menu by short-pressing **SELECT** while in **IDLE NORMAL** or **IDLE GBW** mode.

- **SELECT** → Enter submenu, Enter setting or **SAVE** when in a setting.
- **START** → go back without saving changes
- **SELECT (long press)** → trigger special functions (e.g., calibration)

Menu 1 – Main Menu

No.	Setting	Description
1	**EXIT MENU**	Exit and return to the IDLE display
2	**Purge Settings**	Opens the purge configuration submenu (Menu 2).
3	**GbW Settings**	Opens the grind-by-weight configuration submenu (Menu 3).
4	**Sleep time**	Time (min) before the controller enters deep sleep. Set 0 to disable sleep.
5	**Brightness**	Adjust display brightness (1–255).
6	**LED max Brightness**	Adjust button LED brightness (0–100 %). Set 0 to turn off button LED.
7	**Invert scrolling**	Invert menu scroll direction (CW ↔ CCW).
8	**Max RPM**	Set maximum motor speed limit.
9	**Min RPM**	Set minimum motor speed limit
10	**Motor torque**	Adjust motor torque (%)
11	**Motor ramp**	Control motor rigidity level (level 0–12). A higher rigidity can cause noise, vibration and fast spin up but will keep a tighter RPM. You will need to reboot the servo drive (big PSU box) to activate the new setting.
12	**Calibrate**	Run the auto-purge calibration routine. Long press **SELECT** to begin.

Menu 2 – Purge Settings

No.	Setting	Description
1	**BACK**	Return to Menu 1.
2	**Auto purge enabled**	Enable or disable automatic purge.
3	**Purge /w button**	Run a purge when pressing **START** while the grinding is running.
4	**Purge forward RPM**	Forward purge speed (RPM). Set 0 to disable.
5	**Reverse rotation**	Reverse purge speed (RPM). Set 0 to disable.
6	**Purge delay**	Delay (ms) between end of grind (last bean ground) and purge start.
7	**Purge duration**	Duration (ms) of each purge phase
8	**Purge 0 RPM time**	Time (ms) motor goes to 0 rpm before purging. 0 = immediately spin up (can cause issues with reverse purge atm)
9	**Purge stabilize time**	Stabilization delay after starting or adjusting rpm (ms) before detection triggers.
10	**Purge frames HIGH**	Frame count for high-speed (> 500 RPM) detection.
11	**Purge frames LOW**	Frame count for low-speed (< 500 RPM) detection.
12	**Purge Scalar HIGH**	Sensitivity adjustment (%) for high-speed detection.
13	**Purge Scalar LOW**	Sensitivity adjustment (%) for low-speed detection.
14	**Auto off time**	Time (s) untill the grinder stops if no grinding is detected.

Menu 3 – GbW Settings (Grind-by-Weight)

No.	Setting	Description
1	**BACK**	Return to Menu 1.
2	**GbW RPM**	Motor speed (RPM) for main GbW grinding phase.
3	**GbW Slow RPM**	RPM used in the final slow approach phase.
4	**GbW Slow Phase t**	Duration (ms) of the slow-grind phase, 0 to disable slow phase
5	**Start delay**	Delay (ms) after pressing **START**, preventing false scale reading due to interaction vibrations.
6	**SpeedModifier**	Auto-learning multiplier, not recommended to manually adjust. (default: 4500)
7	**Time offset**	Auto-learning correction value (ms) (default: 200)
8	**Save scale**	Reconnect only to the currently paired scale to prevent unwanted pairing to other scales.

NOTES

- Use caution when changing max RPM, torque, or ramp settings, as these affect motor load and grinder behavior. Reasonably safe limits have been implemented however use at your own risk.
- I am in no way liable for any issues or harm caused by this project
- Future updates will be communicated over EAF discord
- Currently the release version for JMC changes a setting which makes it no longer work with the original controlbox. To fix this, you'll need to manually change a parameter on the servo drive, this controller runs the servo in "Torque mode" allowing a torque limit as a safety feature. The original box runs it in "speed mode" which will push the servo into its (very high) limits indefinetely even when something inside the grinder is blocked. This parameter is P01-01 and must be changed from 2 (torque) back to 1 (speed) to have the original remote controlbox working again.

RESALE IN ANY WAY OR FORM IS NOT ALLOWED WITHOUT EXPLICIT WRITTEN PERMISSION